3.2: Getting Started with Arduino

This module provides an introduction to the Arduino platform, focusing on the Arduino UNO board, the Arduino IDE, and the basic C/C++ syntax used for programming Arduino.

Arduino Platform Overview (UNO Board)

The Arduino UNO is an open-source microcontroller board based on the ATmega328P microcontroller. It is one of the most popular boards in the Arduino family, and its robustness and extensive documentation make it an excellent choice for beginners in electronics and programming.

Key Features of the Arduino UNO R3

Feature	Description
Microcontroller	ATmega328P
Operating Voltage	5V
Input Voltage (recommended)	7-12V
Input Voltage (limits)	6-20V
Digital I/O Pins	14 (of which 6 provide PWM output)
Analog Input Pins	6
DC Current per I/O Pin	20 mA
DC Current for 3.3V Pin	50 mA
Flash Memory	32 KB (ATmega328P) of which 0.5 KB used by bootloader
SRAM	2 KB (ATmega328P)
EEPROM	1 KB (ATmega328P)
Clock Speed	16 MHz

Power

The Arduino UNO can be powered via the USB connection or with an external power supply. The power source is selected automatically.

External (non-USB) power can come either from an AC-to-DC adapter (wall-wart) or a battery. The adapter can be connected by plugging a 2.1mm center-positive plug into the board's power jack. Leads from a battery can be inserted in the Gnd and Vin pin headers of the POWER connector.

Installing the Arduino IDE

The Arduino Integrated Development Environment (IDE) is a cross-platform application (for Windows, macOS, Linux) that is written in functions from C and C++. It

is used to write and upload programs to Arduino compatible boards, but also, with the help of 3rd party cores, other vendor development boards.

Installation Steps

1. **Download the Arduino IDE:** Navigate to the official Arduino website at https://www.arduino.cc/en/software and download the latest version of the Arduino IDE for your operating system.
2. **Install the IDE:**
   - **Windows:** Run the downloaded `.exe` file and follow the on-screen instructions.
   - **macOS:** Drag the downloaded `.dmg` file to your Applications folder.
   - **Linux:** Make the downloaded AppImage file executable and run it.
3. **Launch the Arduino IDE:** Once the installation is complete, open the Arduino IDE.

Basic C/C++ Syntax

Arduino programs are written in a simplified version of C/C++. Every Arduino program has two essential functions: `setup()` and `loop()`.

`setup()`

The `setup()` function is called once when a sketch starts. Use it to initialize variables, pin modes, start using libraries, etc. The `setup()` function will only run once, after each powerup or reset of the Arduino board.

```
void setup() {
  // put your setup code here, to run once:
  pinMode(13, OUTPUT); // Initialize digital pin 13 as an output.
}
```

`loop()`

After creating a `setup()` function, which initializes and sets the initial values, the `loop()` function does precisely what its name suggests, and loops consecutively, allowing your program to change and respond. Use it to actively control the Arduino board.

```
void loop() {
  // put your main code here, to run repeatedly:
  digitalWrite(13, HIGH);   // Turn the LED on (HIGH is the voltage level)
  delay(1000);              // Wait for a second
  digitalWrite(13, LOW);    // Turn the LED off by making the voltage LOW
  delay(1000);              // Wait for a second
}
```

Practice Problems

1. **Blinking LED:** Write a program to make the built-in LED on the Arduino UNO blink with an interval of 2 seconds.
2. **Serial Monitor Output:** Write a program that prints “Hello, Arduino!” to the Serial Monitor every second.

Solutions

1. **Blinking LED:**

```
void setup() {
  pinMode(13, OUTPUT);
}

void loop() {
  digitalWrite(13, HIGH);
  delay(2000);
  digitalWrite(13, LOW);
  delay(2000);
}
```

1. **Serial Monitor Output:**

```
void setup() {
  Serial.begin(9600);
}

void loop() {
  Serial.println("Hello, Arduino!");
  delay(1000);
}
```

References

1. Arduino UNO R3 Documentation
2. Download and install Arduino IDE
3. Arduino `setup()` and `loop()` Functions Explained